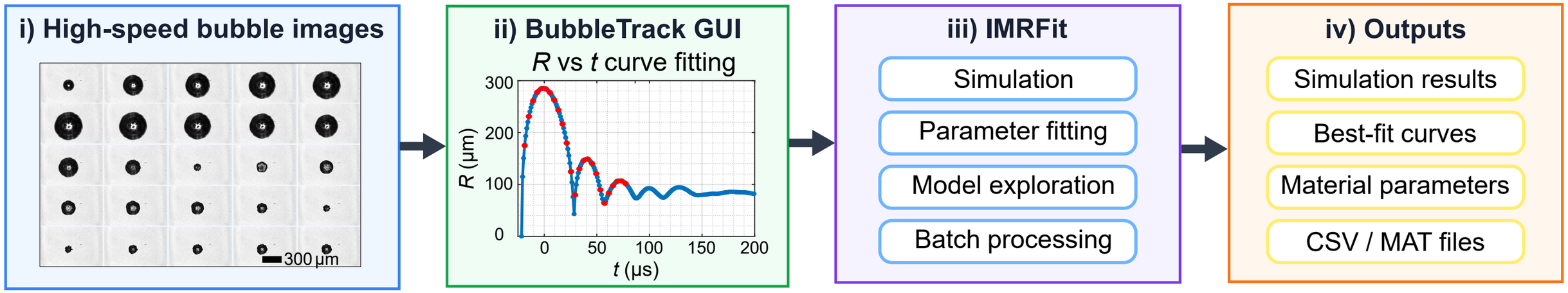
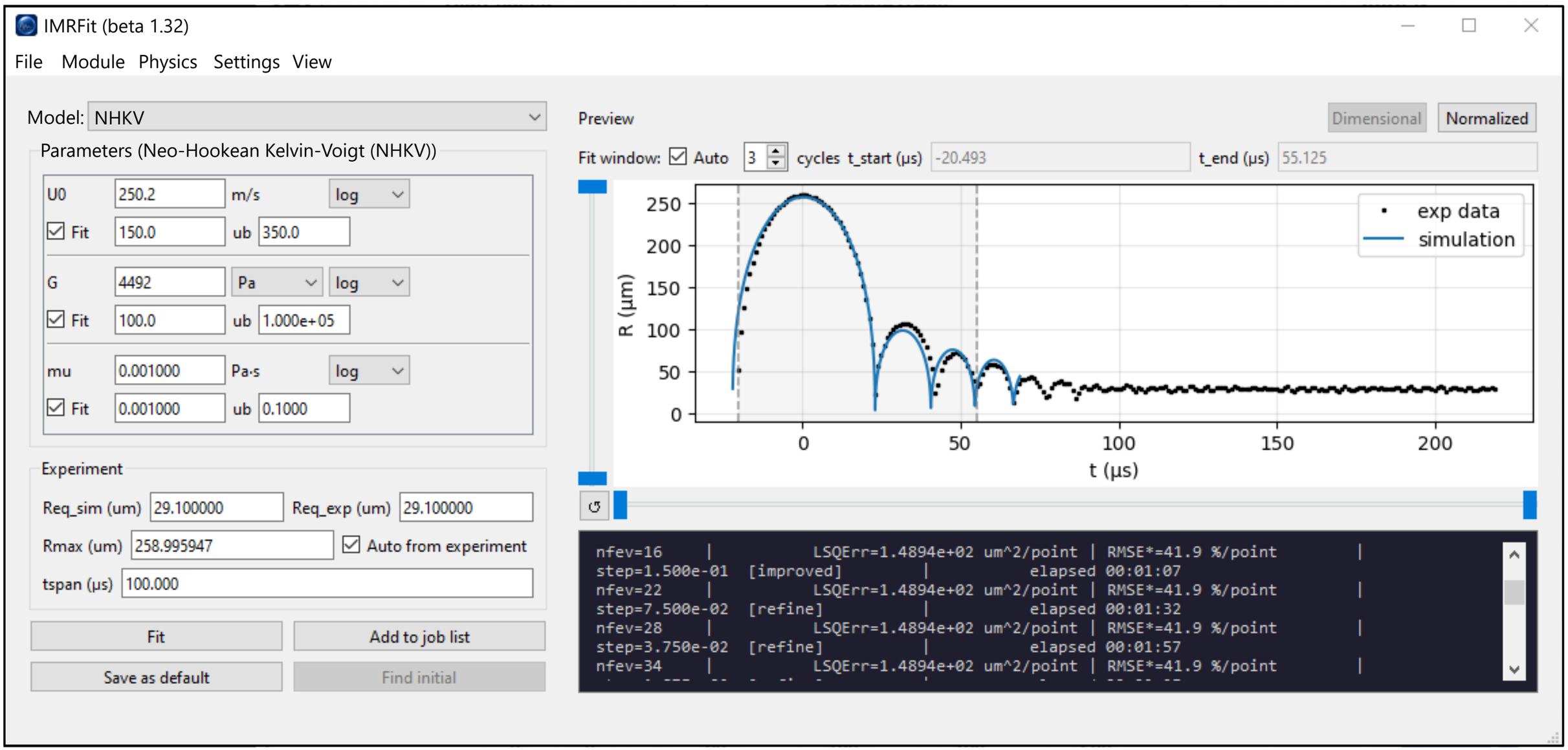
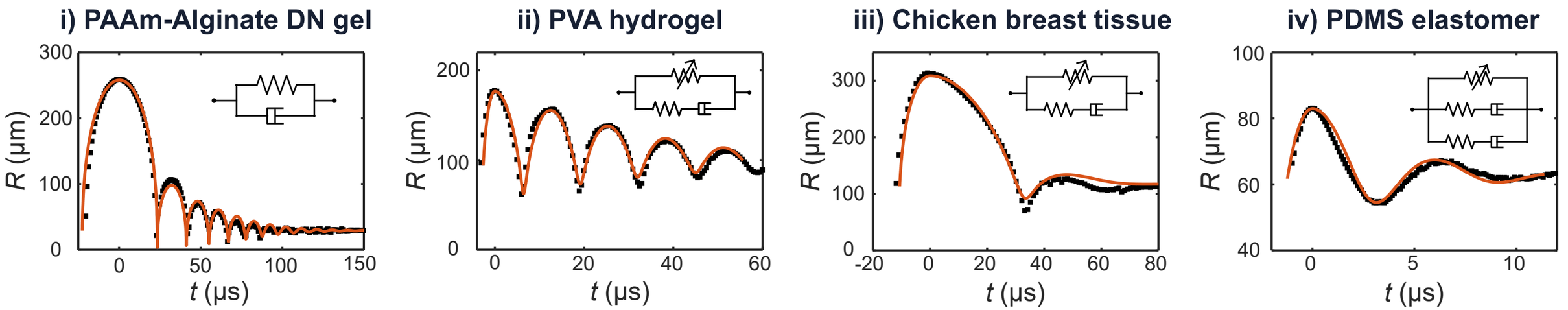
A


i) High-speed bubble images
300 μm
ii) BubbleTrack GUI
R vs t curve fitting
R (μm)
t (μs)
iii) IMRFit
Simulation
Parameter fitting
Model exploration
Batch processing
iv) Outputs
Simulation results
Best-fit curves
Material parameters
CSV / MAT files

B


IMRFit (beta 1.32)
File Module Physics Settings View
Model: NHKV
Parameters (Neo-Hookean Kelvin-Voigt (NHKV))
U0 250.2 m/s log
Fit 150.0 ub 350.0
G 4492 Pa log
Fit 100.0 ub 1.000e+05
mu 0.001000 Pa·s log
Fit 0.001000 ub 0.1000
Experiment
Req_sim (um) 29.100000
Req_exp (um) 29.100000
Rmax (um) 258.995947
Auto from experiment
tspan (μs) 100.000
Fit
Add to job list
Save as default
Find initial
Preview
Dimensional
Normalized
Fit window: Auto 3 cycles t_start (μs) -20.493 t_end (μs) 55.125
exp data
simulation
R (μm)
t (μs)
nfev=16 | LSQErr=1.4894e+02 um^2/point | RMSE*=41.9 %/point |
step=1.500e-01 [improved] | elapsed 00:01:07
nfev=22 | LSQErr=1.4894e+02 um^2/point | RMSE*=41.9 %/point |
step=7.500e-02 [refine] | elapsed 00:01:32
nfev=28 | LSQErr=1.4894e+02 um^2/point | RMSE*=41.9 %/point |
step=3.750e-02 [refine] | elapsed 00:01:57
nfev=34 | LSQErr=1.4894e+02 um^2/point | RMSE*=41.9 %/point |

C


i) PAAm-Alginate DN gel
ii) PVA hydrogel
iii) Chicken breast tissue
iv) PDMS elastomer
R (μm)
t (μs)